Linda Ebbert[1], Julian Varghese[1], Sarah Sandmann[1]
[1]Institute of Medical Informatics, University of Münster, Germany

Institut für Medizinische Informatik

tRuffle: an R/Shiny app for exploring unique DEGs in scRNA-seq clusters

Introduction

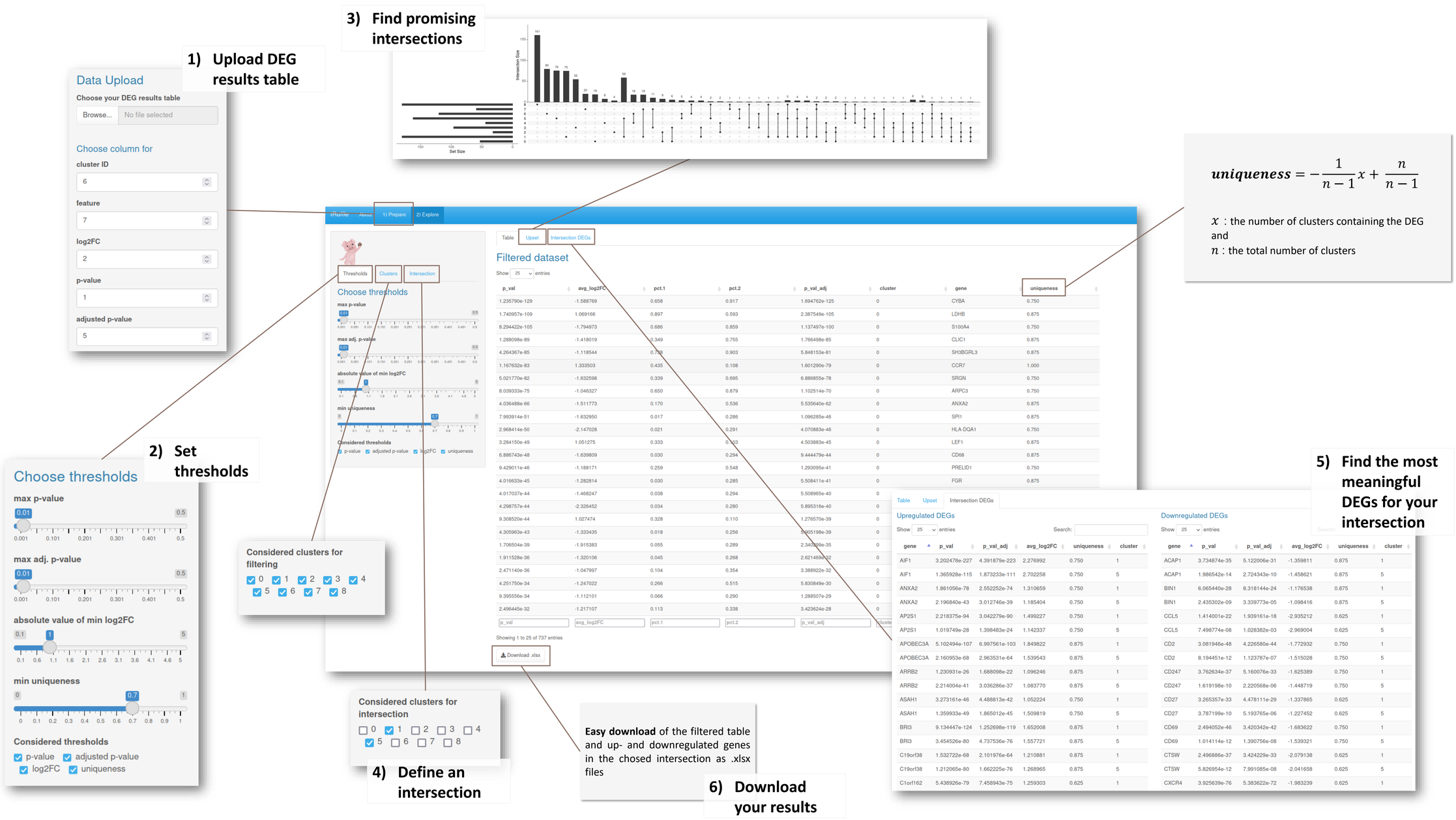
Single-cell RNA-sequencing (scRNA-seq) represents a powerful method to investigate differential gene expression patterns among distinct cell types. Although the majority of identified differentially expressed genes (DEGs) are based on biological differences, there are still those that are due to technical noise and are not specific to a cluster or small cluster group [1].

Methods

Based on the DEG occurrence among all clusters, a uniqueness-score is calculated for every gene. Subsequently, this score operates as an additional evaluation parameter by excluding genes that are found to be differentially expressed in many clusters and are thus unspecific.
A score of 0 would represent DEGs found in all given clusters, while genes differentially expressed in one single cluster are scored as 1.

Results

tRuffle is a user-friendly R/Shiny app, which does not only help to exclude systematic artifacts, but in addition, allows to easily recognize the most meaningful DEGs for each scRNA-seq cluster.



Discussion & Conclusion

Individual thresholds enable the user to adjust for varying levels of background noise and to explore only those genes which show cluster-specific expression. The combination of tabular and graphical output makes tRuffle a convenient, fast and adaptable tool to generate valid biological hypotheses from scRNA-seq data.

Literature/References

[1] Tung, P. Y., Blischak, J. D., Hsiao, C. J., Knowles, D. A., Burnett, J. E., Pritchard, J. K., & Gilad, Y. (2017). Batch effects and the effective design of single-cell gene expression studies. Scientific reports, 7, 39921.

UKM Universitätsklinikum Münster

WWU MÜNSTER

medizinische fakultät Westfälische Wilhelms-Universität Münster